

Totally Integrated Automation Portal		
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REACTION_VESSEL / PLC_1 [CPU 1215C DC/DC/DC] / Program blocks

Main [OB1]

Main Properties

General

Name	Main	Number	1	Type	OB
Language	LAD	Numbering	Automatic		

Information

Title	REACTION_VESSEL	Author	SHREYASH_GHADGE	Comment	I-1 – Start I-2 – Low Level I-3 – High Level I-4 – Low Pressure I-5 – High Pressure I-6 - Low Temperature I-9 – High Temperature I-10- Stop Q-1 – Inlet Valve Q-2 – Outlet Valve ,Q-3 – Cooling Water Pump (for reducing Temp.)Q-4 – Safety Valve (for releasing excess Pressure) Q-5 –Heater,Q-6 – Stirrer,Q-7 – Normal Conditions (indication),Q-8 – Alarm (Indication) When Start button is pressed, if the level is Low, Start Inlet Valve. When Temperature or Pressure is Low, Give an Alarm and Start Heater and Stirrer. If Pressure is High, give an Alarm, Stop Heater, Open Safety Valve. If Temp.is High, give Alarm, stop Heater, start Cooling Water Pump. When there is no Alarm, Give Normal Indication. When High Level is attained, stop Inlet Valve, stop Heater & Stirrer When it is High Level and Normal Condition, Start Outlet Valve. When Low Level is reached, Stop Outlet Valve, start inlet valve And repeat the process.
Family		Version	0.1	User-defined ID	

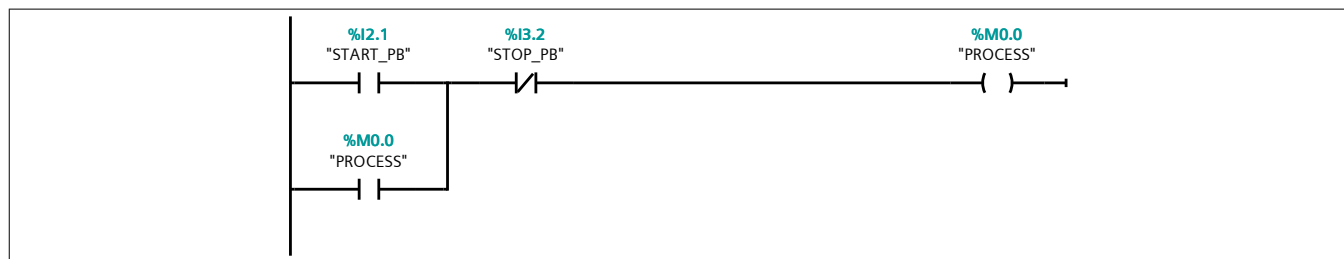
Main

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB

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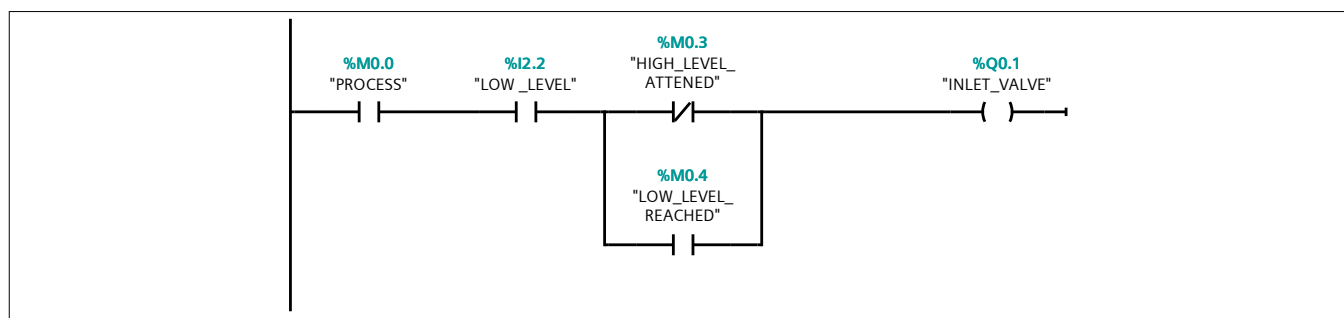
Name	Data type	Default value	Comment
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1: I-1 – Start ,I-10- Stop



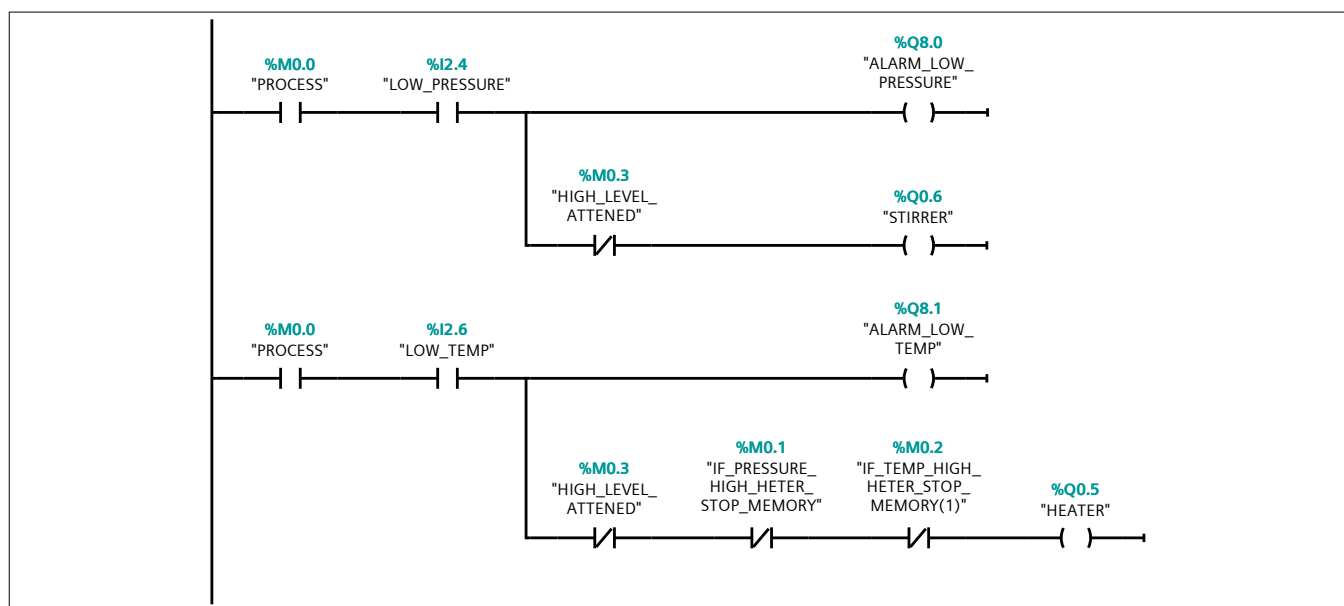
Network 2: Start Inlet Valve

When Start button is pressed, if the level is Low,
Start Inlet Valve



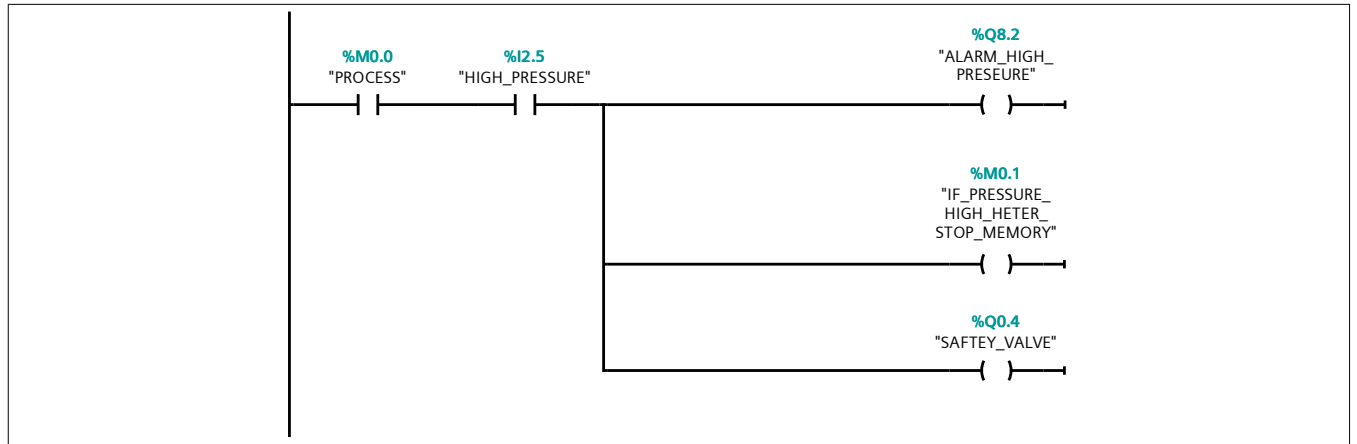
Network 3: Start Heater and Stirrer.

When Temperature or Pressure is
Low, Give an Alarm and Start Heater and Stirrer.



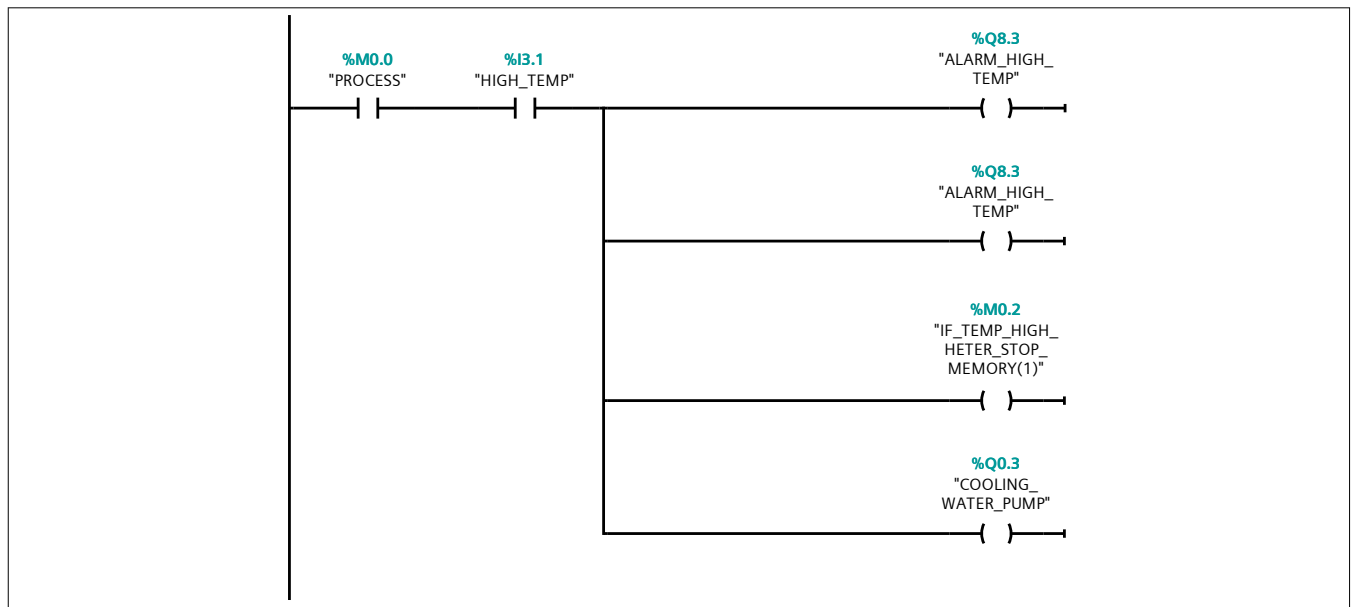
Network 4: Open Safety Valve.

If Pressure is High, give an Alarm, Stop Heater, Open Safety Valve.



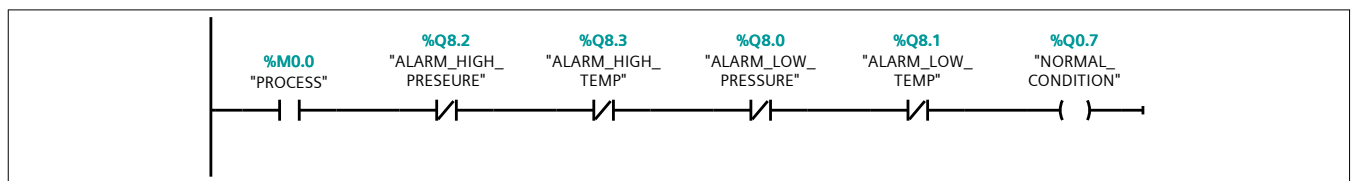
Network 5: start Cooling Water Pump.

If Temp.is High, give Alarm, stop Heater, start Cooling Water Pump.



Network 6: Normal Indication.

When there is no Alarm, Give Normal Indication.



Network 7: stop Inlet Valve, stop Heater & Stirrer

When High Level is attained, stop Inlet Valve, stop

Totally Integrated Automation Portal		
Heater & Stirrer		
%M0.0 "PROCESS" %I2.3 "HIGH_LEVEL" %M0.3 "HIGH_LEVEL_ ATTENED"		
Network 8: Start Outlet Valve.		
When it is High Level and Normal Condition, Start Outlet Valve.		
%M0.0 "PROCESS" %I2.3 "HIGH_LEVEL" %Q0.7 "NORMAL_ CONDITION" %M0.4 "LOW_LEVEL_ REACHED" %Q0.2 "OUTLET_VALVE"		
Network 9:		
When Low Level is reached, Stop Outlet Valve, start inlet valve And repeat the process.		
%M0.0 "PROCESS" %I2.2 "LOW_LEVEL" %M0.4 "LOW_LEVEL_ REACHED"		